

STOWLINE · 3D CONTAINER STOWAGE PLANNER

Catch stowage mistakes at the moment of placement

For vessel planners at shipping lines and terminals who plan a port call under time pressure

THE PROBLEM

Most stowage mistakes are found late, when they are expensive to fix

What goes wrong	What it costs
Overstow: an early-port box under a later-port box	Extra crane moves and a longer port stay
A stack heavier than its limit	Safety risk, and the vessel can reject the plan
A reefer in a slot with no power plug	The cargo can spoil
Dangerous goods next to an incompatible class	Safety and compliance risk
Weight badly spread along or across the ship	Poor trim, list or stability, and last-minute replanning

THE PROBLEM

Why today's tools let mistakes through

1

One bay at a time

A 2D bay plan shows one bay at a time, so problems that cross bays or ports are hard to see.

2

Checks run afterwards

Rule checks often run as a separate step after planning, not while the planner works.

3

Stability shows up last

The effect of one move on stability is not visible until the whole plan is calculated.

THE SOLUTION

One workspace for the whole ship

- The whole ship in 3D and one bay in 2D, always in sync
- Every rule checked before the container is dropped
- Stability visible while planning, not after
- A clear handoff: Draft, In review, Approved



Workspace screenshot: 3D view, bay view, load list, stability strip
or [browse files](#)

HOW IT WORKS

From load list to approved plan in one flow



Import

The load list is checked row by row. Rejected rows show the reason.

Plan

Drag, click or use the keyboard. Valid and invalid slots show before the drop.

Fix

Each violation comes with a reason. Apply fix makes the suggested move.

Review

The senior planner approves, or returns the plan with a comment.

Hand over

The terminal exports the approved, locked plan.

After sailing, the departure condition becomes the arrival condition at the next port, so each port call starts from the real state of the ship.

RULE CHECKS

Six rules checked on every move

Error

Stack weight

90.0 t on deck, 210.0 t in the hold

Error

Reefer power

Reefers only in slots with a plug

Error

Dangerous goods

Incompatible classes kept apart, including across bays

Error

Overstow

Shows the number of restow moves it would cause

Error

20ft and 40ft

No 20ft on a 40ft, no 40ft on a half-empty pair

Warning

Heavy over light

More than 10.0 t heavier than the box below

A drop that causes an error is refused with the reason. Every violation offers a suggested fix when one exists.

STABILITY

Stability visible during every drag

GM

1.84 m

OK · limit 1.20 m

Trim

0.62 m

OK · limit 1.50 m

List

0.4°

Check · limit 2.0°

Hull strength

78%

OK · limit 100%

The stability strip is always on screen and previews the change in GM, trim and list while a container is held over a slot.

Each value reads OK, Check or Limit, so the planner sees a caution before a limit is reached.

Values shown are from the sample vessel MV Nusantara Pioneer. Version 1 stability is a simplified estimate, not a certified loading computer.

WHO IT SERVES

One plan, four roles

Vessel planner

Goal

Finish a safe plan before the terminal cut-off

Senior planner

Goal

Approve only plans that are safe and cheap to work

Terminal planner

Goal

Know where every container goes before loading starts

Chief officer

Goal

Confirm the ship stays stable and within strength limits

CONTROL

Approval, versions and undo built in

Draft

Send for review →

In review

Approve, zero errors →

Approved

No unsafe approvals

Approve stays disabled while any error remains.
An approved plan is locked.

No overwritten work

Every save is a new version. If a colleague saved first, the save is refused and the moves are kept.

Every change reversible

Undo and redo work for every move, from the keyboard or from each result message.

BUILD TARGETS

Fast enough for an expert, open to every planner

<100 ms

from hover to a valid or invalid mark

55 fps

in 3D with 10,000 containers on a mid-range laptop

<2.5 s

until the plans list is usable on fast 4G

6 of 6

rule types flagged before save

0 mouse

needed from load list to saved plan

**WCAG 2.2
AA**

with a keyboard and screen reader path for the 3D view

BUSINESS IMPACT

What a rollout would measure



Restow moves per port call

Fewer extra crane moves and shorter port stays



Violations found after approval

Errors are caught before the plan leaves the planner



Time from load list to approved plan

Less replanning at the last minute

Baselines are not measured yet. A pilot would set them.

SCOPE

Version 1 today and the path to production

In version 1

The full process, from load list to approved handoff

17 features, 6 live rule checks, stability estimate, port playback

Desktop web app at 1440 × 900 and 1920 × 1080

JSON import and export, realistic sample data

Needed for live operations

EDI connection (BAPLIE, COPRAR, MOVINS)

A real backend in place of the browser mock API

A certified loading computer and the full IMDG table

Crane sequencing, lashing and line-of-sight checks

Next steps

- 1 A live demo on the sample port call in Singapore
- 2 Feedback sessions with your working planners
- 3 A pilot on one service to set the baselines for restows and planning time